CLUSTERING & CLASSIFYING CREDIT CARD CUSTOMERS

STAT 101 Final Project
Part 1

PREPARED BY:

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Executive Summary

The number of clusters was determined based on whether an additional cluster identified a group with characteristics significantly different from another cluster. Five clusters were identified. Cluster 1 is the largest. Clusters 2-5 share all but one characteristic with Cluster 1. Cluster 2 has a higher number of credit cards. Cluster 3 has a lower average balance. Cluster 4 has a higher number of open bank accounts. Cluster 5 owns more houses.

A random bootstrap forest model was created to predict whether an individual would accept the credit card offer. Due to the unbalanced nature of the data set a model created without any changes to the data completely missed the small subset of individuals who accepted the credit card offer. As a result, synthetic data mimicking the minority group was inserted into the data to balance the groups. The resulting model correctly predicted whether an individual would accept the offer 94.3% of the time when tested against new data.

Details of Analysis and Recommendations

Data Preparation:

Addressing Missing Values:

24 rows contained no data for “Q1 Balance,” “Q2 Balance,” “Q3 Balance,” “Q4 Balance,” and “Average Balance.” All 24 rows were excluded. The excluded data included 2 individuals who accepted the offer and 22 individuals who rejected the offer.

Checking Outliers:

Significant yet reasonable outliers identified in “# Bank Accounts Open,” “# Credit Cards Held,” “Household Size,” “Q1 Balance,” “Q2 Balance,” “Q3 Balance,” “Q4 Balance,” and “Average Balance.”

Uncorrelated Data Removal:

“Customer Number” was excluded from all analysis processes.

Correlated Data Removal:

Since the “Average Balance” and Q1, Q2, Q3, and Q4 balances are correlated, and use of correlated parameters can introduce excess noise into the data set, it makes sense in some cases to only use the “Average Balance” column.

For Clustering Analysis: Only “Average Balance” used.
For Prediction Model: All balance related columns used.

Exploratory Data Analysis:

Strongest Correlations:

"Average Balance" & "Q2 Balance" (.8003)
"Average Balance" & "Q3 Balance" (.7527)

	Average Balance	Q1 Balance	Q2 Balance	Q3 Balance	Q4 Balance
Average Balance	1.0000	0.4815	0.8003	0.7527	0.5724
Q1 Balance	0.4815	1.0000	0.4831	-0.0610	-0.2373
Q2 Balance	0.8003	0.4831	1.0000	0.4877	0.1690
Q3 Balance	0.7527	-0.0610	0.4877	1.0000	0.5653
Q4 Balance	0.5724	-0.2373	0.1690	0.5653	1.0000

Results of Customer Segmentation Analysis:

Considering Different Numbers of Clusters:

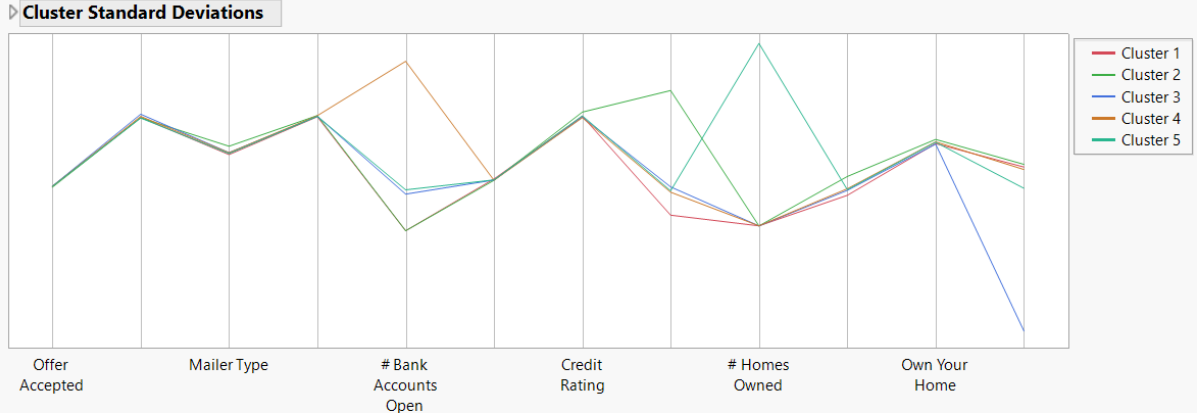
The addition of a fifth cluster adds a unique group who hold a higher number of credit cards. Additionally, the fifth cluster is similar in size to the other clusters. The addition of a sixth cluster adds a smaller group with no significant differences from Cluster 4. Since the addition of a fifth cluster added a group with at least one unique characteristic, and the addition of a sixth cluster did not, five clusters were used.

Cluster Breakdown:

Cluster 1 is the largest cluster and shares many characteristics with the other clusters. Clusters 2-5 share all but one characteristic with Cluster 1.
Cluster 2 includes individuals who hold a higher number of credit cards (with a mean of 2.98).
Cluster 3 includes individuals who have a lower average balance (with a mean of 276.37).
Cluster 4 includes individuals who have a higher number of open bank accounts (with a mean of 2.08).
Cluster 5 includes individuals who own a greater number of houses (with a mean of 2.05).

Cluster	Count	Offer Accepted	Reward	Mailer Type	Income Level	# Bank Accounts Open	Overdraft Protection	Credit Rating	# Credit Cards Held	# Homes Owned	Household Size	Own Your Home	Average Balance
1	7590	0.07016	1.80075	0.75144	1.70864	1.00000	0.20510	1.82735	1.64	1.00	3.43	0.95	1052.08
2	1836	0.06726	1.79588	0.80793	1.71975	1.00000	0.19830	1.87332	2.98	1.00	3.71	0.98	1064.64
3	2121	0.07088	1.83628	0.76458	1.71313	1.23338	0.20005	1.81773	1.94	1.00	3.50	0.95	276.37
4	2958	0.07034	1.81231	0.76401	1.71896	2.08080	0.19857	1.81119	1.89	1.00	3.53	0.96	1040.35
5	3471	0.07000	1.79364	0.75831	1.71196	1.26073	0.20055	1.83214	1.90	2.05	3.51	0.96	951.66

Cluster Standard Deviations



Recommendations:

As the largest Cluster, general credit card offers should be designed to fit the needs of individuals within Cluster 1. Alternative offers could be targeted towards individuals with more credit cards, individuals who have a smaller balance, individuals who have more bank accounts, or individuals who own more houses. Offers that try to appeal to a specific audience by altering Reward, Mailer Type, Income Level, Overdraft Protection, Household Size, or Home Ownership may be less effective unless used in conjunction with offers to target the individuals within clusters 2-5.

Results of Predictive Model Analysis:

Modifying the Dataset:

Creating a model to predict offer acceptance is complicated by the unbalanced nature of the data set. The offer accepted column contains a minority class with only 5.7% of individuals within the dataset accepting the offer. To balance the data set a Synthetic Minority Oversampling Technique (SMOTE) was conducted. The JMP Imbalanced Classification Add-In used can be found at https://community.jmp.com/t5/JMP-Add-Ins/Imbalanced-Classification-Add-In/ta-p/304836. Within the add-in, the "SOMTE Observations" technique was used to balance the sizes of the majority and minority groups. After adding synthetic data with SMOTE, 39.7% of individuals accepted the offer.

Creating the Model:

Before balancing the dataset, a validation column was created containing a training proportion of .55, validation proportion of .3, and a test proportion of .15. The SMOTE process increased the proportion of the minority for the training and validation groups but did not change the proportion for the test group. A bootstrap random forest classification model was then created using that validation column.

Bootstrap Forest for Offer Accepted

Specifications

Target	Offer Accepted	Training Rows:	15328
Validation Column:	Validation	Validation Rows:	10225
		Test Rows:	2633
Number of Trees in the Forest:	100	Number of Terms:	15
Number of Terms Sampled per Split:	11	Bootstrap Samples:	15328
		Minimum Splits per Tree:	10
		Minimum Size Split:	28

Overall Statistics

Measure	Training	Validation	Test	Definition
Entropy RSquare	0.6855	0.4489	0.5063	1-Loglike(model)/Loglike(0)
Generalized RSquare	0.8146	0.6168	0.6155	(1-(L(0)/L(model))^(2/n))/(1-L(0)^(2/n))
Mean -Log p	0.2118	0.3798	0.2198	∑ -Log(ρ[j])/n
RASE	0.2354	0.3441	0.2354	√ ∑(y[j]-ρ[j])²/n
Mean Abs Dev	0.1681	0.2567	0.1541	∑ |y[j]-ρ[j]|/n
Misclassification Rate	0.0595	0.1589	0.0539	∑ (ρ[j]≠ρMax)/n
N	15328	10225	2633	n

Confusion Matrix

Training

Actual Offer Accepted	Predicted Count No	Predicted Count Yes
No	9054	129
Yes	783	5362

Actual Offer Accepted	Predicted Rate No	Predicted Rate Yes
No	0.986	0.014
Yes	0.127	0.873

Validation

Actual Offer Accepted	Predicted Count No	Predicted Count Yes
No	5432	137
Yes	1488	3168

Actual Offer Accepted	Predicted Rate No	Predicted Rate Yes
No	0.975	0.025
Yes	0.320	0.680

Test

Actual Offer Accepted	Predicted Count No	Predicted Count Yes
No	2200	3
Yes	139	291

Actual Offer Accepted	Predicted Rate No	Predicted Rate Yes
No	0.999	0.001
Yes	0.323	0.677

Other Method Considerations:

A Neural network through JMP Classification was tested using the validation column with the test group. The model performed like the Bootstrap Forest on the validation group, but worse on the test group. As a result, the Bootstrap Forest model is more effective.

Neural

Validation Column: Validation

Model Launch

Model NTanH(3)

Training

Offer Accepted

Measures	Value
Generalized RSquare	0.2957049
Entropy RSquare	0.183345
RASE	0.4335745
Mean Abs Dev	0.3804934
Misclassification Rate	0.2934499
-LogLikelihood	8429.0985
Sum Freq	15328

Confusion Matrix

Actual Offer Accepted	Predicted Count No	Yes
No	7202	1981
Yes	2517	3628

Confusion Rates

Actual Offer Accepted	Predicted Rate No	Yes
No	0.784	0.216
Yes	0.410	0.590

Validation

Offer Accepted

Measures	Value
Generalized RSquare	0.277988
Entropy RSquare	0.1691276
RASE	0.4430666
Mean Abs Dev	0.3937493
Misclassification Rate	0.3101222
-LogLikelihood	5854.8371
Sum Freq	10225

Confusion Matrix

Actual Offer Accepted	Predicted Count No	Yes
No	3896	1673
Yes	1498	3158

Confusion Rates

Actual Offer Accepted	Predicted Rate No	Yes
No	0.700	0.300
Yes	0.322	0.678

Test

Offer Accepted

Measures	Value
Generalized RSquare	0.0678835
Entropy RSquare	0.0458707
RASE	0.3639841
Mean Abs Dev	0.2819215
Misclassification Rate	0.1644512
-LogLikelihood	1118.243
Sum Freq	2633

Confusion Matrix

Actual Offer Accepted	Predicted Count No	Yes
No	2159	44
Yes	389	41

Confusion Rates

Actual Offer Accepted	Predicted Rate No	Yes
No	0.980	0.020
Yes	0.905	0.095

Recommendations:

The overall misclassification rate is 5.7%. However, the model misses about one third of the individuals who would accept the offer. With more data, particularly more accepted offers, the model may be improved. Additionally, more advanced model creation techniques, besides the baseline methods included in JMP software, may be helpful.

Suggestions for Future Work:

The current credit card offers should be examined to determine whether they fit the needs of individuals within Cluster 1. The average attributes of Cluster 1 are shown below.

Cluster	Count	Offer Accepted	Reward	Mailer Type	Income Level	# Bank Accounts Open	Overdraft Protection	Credit Rating	# Credit Cards Held	# Homes Owned	Household Size	Own Your Home	Average Balance
1	7590	0.07016	1.80075	0.75144	1.70864	1.00000	0.20510	1.82735	1.64	1.00	3.43	0.95	1052.08

Additionally, the current credit card offers should be examined for compatibility with individuals who have a higher number of credit cards, a smaller balance, a higher number of bank accounts, or own a greater number of houses. If current offers are not suitable for those groups, they should be modified, or new offers should be created.

SPAM VS HAM

STAT 101 Final Project
Part 2

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Model Creation

The Text Explorer in JMP with all terms stemmed was used to analyze the messages. The document term matrix of the text analysis was saved as an IDF Matrix. A Bootstrap Forest model with a validation proportion of .3 was created.

Bootstrap Forest for Spam or Ham?

Specifications

Target	Spam or Ham?	Training Rows:	3865
		Validation Rows:	1707
Number of Trees in the Forest:	100	Test Rows:	0
Number of Terms Sampled per Split:	452	Number of Terms:	966
		Bootstrap Samples:	3865
		Minimum Splits per Tree:	10
		Minimum Size Split:	5

Overall Statistics

Measure	Training	Validation	Definition
Entropy RSquare	0.8295	0.7326	1-Loglike(model)/Loglike(0)
Generalized RSquare	0.8809	0.8016	(1-(L(0)/L(model))^(2/n))/(1-L(0)^(2/n))
Mean -Log p	0.0684	0.1009	∑ -Log(ρ[j])/n
RASE	0.1326	0.1604	√ ∑(y[j]-ρ[j])²/n
Mean Abs Dev	0.0508	0.0613	∑ |y[j]-ρ[j]|/n
Misclassification Rate	0.0217	0.0299	∑ (ρ[j]≠ρMax)/n
N	3865	1707	n

Confusion Matrix

Training

Actual Spam or Ham?	Predicted Count ham	Predicted Count spam
ham	3321	11
spam	73	460

Actual Spam or Ham?	Predicted Rate ham	Predicted Rate spam
ham	0.997	0.003
spam	0.137	0.863

Validation

Actual Spam or Ham?	Predicted Count ham	Predicted Count spam
ham	1487	6
spam	45	169

Actual Spam or Ham?	Predicted Rate ham	Predicted Rate spam
ham	0.996	0.004
spam	0.210	0.790

Misclassification Rate Analysis

Validation Misclassification Rate:

The validation misclassification rate is 3.08%

(15+27) / (196+195) = .0308 = 3.08%

Highest Percentage of Misclassifications:

Spam had a higher misclassification rate at 21.0% compared to .4% for Ham.

ETHICS ANALYSIS

STAT 101 Final Project
Part 3

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Skin cancer data is difficult to find, likely due to privacy reasons, so I am working with the following assumption.

ASSUMPTION: Dermatologists diagnose about the same number of skin abnormalities as cancerous and non-cancerous (allows for balanced comparison of percentages).

CONSIDERATION 1:

Would it be ethical to entirely remove physicians from the diagnosis procedure and leave the diagnosis entirely to data science methods?

Confusion matrix for dermatologists diagnosing skin cancer:

Actual	Diagnosed as Cancerous	Diagnosed as Non-Cancerous
Cancerous	88.9%	11.1%
Non-Cancerous	24.3%	75.7%

Confusion matrix for data science methods:

Actual	Diagnosed as Cancerous	Diagnosed as Non-Cancerous
Cancerous	90.9%	9.1%
Non-Cancerous	44.3%	55.7%

It is ethical to switch to only Data Science methods if more lives are saved. When switching, an additional 2% of cases are correctly classified as cancerous. This increase in correct classification results in more individuals with cancer getting treated, saving lives. Although an additional 20% of cases are classified as cancerous when they are benign, the cancer treatment without cancer is likely less deadly than the cancer without any treatment. The extent to which that is true determines if the switch is ethical. If the mortality rate of the average treatment is less than 1/10th of the mortality rate of the average cancer, more lives would be saved by switching to Data Science methods. In other words, if someone with cancer and no treatment is over ten times more likely to die than someone with no cancer still getting cancer treatment, then it is ethical to switch to only Data Science methods.

CONSIDERATION 2:

Would it be ethical to entirely remove physicians from the diagnosis procedure and leave the diagnosis entirely to data science methods?

Confusion matrix for dermatologists diagnosing skin cancer:

Actual	Diagnosed as Cancerous	Diagnosed as Non-Cancerous
Cancerous	88.9%	11.1%
Non-Cancerous	24.3%	75.7%

Confusion matrix for data science methods:

Actual	Diagnosed as Cancerous	Diagnosed as Non-Cancerous
Cancerous	78.9%	21.1%
Non-Cancerous	4.3%	95.7%

It is ethical to switch to only Data Science methods if more lives are saved. When switching, an additional 10% of cases are incorrectly classified as non-cancerous. This incorrect classification results in more individuals with cancer not getting treated, costing lives. Although an additional 20% of cases are correctly classified as non-cancerous, cancer without any treatment is likely more deadly than the cancer treatment without cancer. If the mortality rate of the average treatment is less than half of the mortality rate of the average cancer, fewer lives would be saved by switching to Data Science methods. In other words, if someone with cancer and no treatment is over twice as likely to die than someone with no cancer still getting cancer treatment, then it is ethical to continue letting dermatologists diagnose skin abnormalities.

CONSIDERATION 3:

How does your opinion on this question compare to your opinion regarding letting automobiles be entirely self-driving?

If a self-driving car is less likely to get into a fatal accident than the average 18-year-old, living in the Midwest, with 2.5 years of experience, then I am perfectly comfortable letting the car drive itself. While everyone might not hold the same opinion, applied to a population, if the self-driving car is less likely to get into a fatal crash than the average individual in the population, it is ethical to switch to entirely self-driving cars. The main challenge is proving that that is or is not the case. The central theme between my opinion on the ethics of self-driving cars and using Data Science to diagnose skin cancer is the idea that whichever option will reduce fatalities is more ethical.